

Customer Churn Analytics Report

Executive Summary

This analysis covers **21 customers** in `customer_churn_data_raw.xlsx`. Six customers have a cancellation date, giving an observed churn rate of **28.57%** and retention of **71.43%**. Their monthly charges total **73.94** in the source data, or **18.68%** of the full sample's monthly charges (395.79). The source does not document the currency or scale of these amounts.

The strongest observed patterns are:

- **Basic** has the highest plan-level churn rate (60%, 3 of 5 customers); **Premium** is lowest (14.29%, 1 of 7).
- **Karnataka** (2 of 2) and **Meghalaya** (2 of 3) have the highest state-level churn rates. State sample sizes are very small, so these rates are directional rather than reliable market estimates.
- All four customers with an explicitly recorded support escalation churned. This is a strong warning signal in this sample, but sparse support coverage and the treatment of missing support records make the full-sample correlation unsuitable as causal evidence.
- The score-based **high-risk** segment contains six customers and all six churned. This is a descriptive result on the same small dataset, not independent validation of churn-score predictive performance.

Dataset and Scope

- **Customer/subscription records:** 21 each, with 21 unique customer IDs.
- **Support records:** 9 complaint rows across 7 customers; two customers have duplicate complaint rows.
- **Churn definition:** `cancellation_date` is present.
- **Analysis unit:** one row per customer after joining the customer, subscription, and deduplicated support data.
- **Time coverage of observed cancellations:** February through November 2024. This file is a small snapshot, not a complete monthly customer-base history.

Data Cleaning

The Python analysis script was run against the workbook and completed with exit code 0. Its cleaning steps were:

1. Read the `db_customer`, `db_subscription`, and `db_support` sheets.
2. Renamed customer `name` to `customer_name`.
3. Dropped `interests` (only 4 of 21 populated) and `pincode` (0 of 21 populated).
4. Normalized gender values from `Men/Women` to `Male/Female`.
5. Filled 3 missing customer countries using the country associated with the same state; no country values remained missing after mapping.
6. Parsed subscription start, renewal, cancellation, and complaint dates as dates.
7. Dropped the support `col_1` and `comment` columns.
8. Counted raw complaint rows per customer, sorted support rows by complaint date, and retained the latest row per customer for the customer-level join. The final join has 21 rows, avoiding duplication from multiple complaints.

Data-quality caveats: the state-to-country fill assumes a state maps consistently to one country. Support history is reduced to the latest complaint for joined attributes, while complaint counts preserve the number of raw rows. The intermediate join printed by the script has 23 rows due to duplicate support records; the final deduplicated join correctly returns 21.

Feature Engineering

- **churn_flg:** 1 when **cancellation_date** is present, otherwise 0.
- **complaint_count:** number of raw support rows per customer, attached to that customer's retained latest support row.
- **churn_risk:** score-based bands: **low** for score < 50, **med** for 50–69, and **high** for score ≥ 70.
- **cancellation_month:** month derived from the cancellation date for monthly churn counts.
- **Escalation indicator:** **Y** is converted to 1 and every other value—including missing support records—to 0 in the script's correlation calculation.

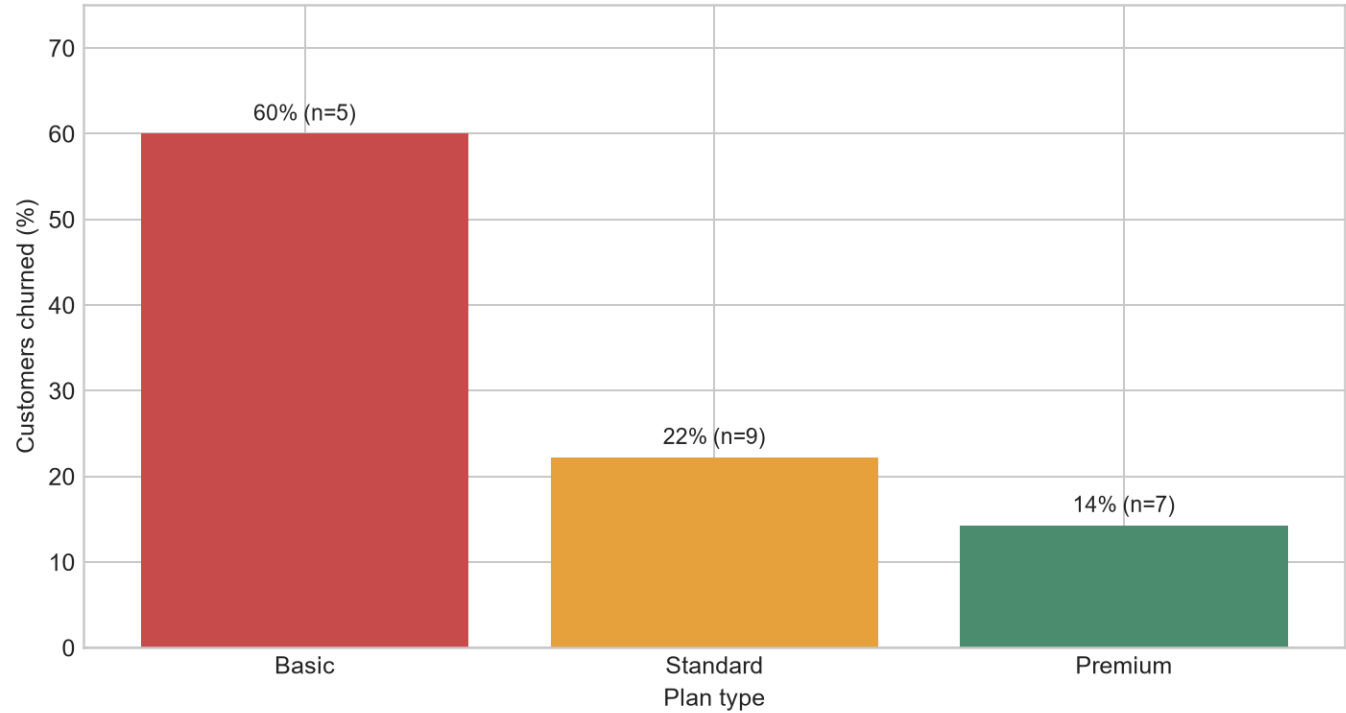
KPI Scorecard

KPI	Result	Definition / interpretation
Customers	21	Final one-row-per-customer dataset
Churned customers	6	Cancellation date present
Churn rate	28.57%	6 / 21
Retention rate	71.43%	1 – churn rate
Total monthly charges	395.79	Sum of monthly_charges in the sample
Revenue at risk	73.94	Monthly charges of the 6 churned customers; source currency/scale unknown
Revenue at risk share	18.68%	73.94 / 395.79
Average revenue per user (ARPU)	18.85	Mean monthly charges across all 21 customers
Explicit escalation rate	19.05%	4 Y rows / 21 customers; missing support is not explicitly escalated
Average complaints per customer	0.43	9 raw complaint rows / 21 customers

ARPU here uses all customers in the final sample, including churned customers. Revenue at risk is a **monthly charge run-rate**, not lifetime value or annualized revenue. The script prints “Rs 'K'” beside revenue at risk, but the workbook does not document the currency or scale, so this report does not assume one.

Churn by Plan Type

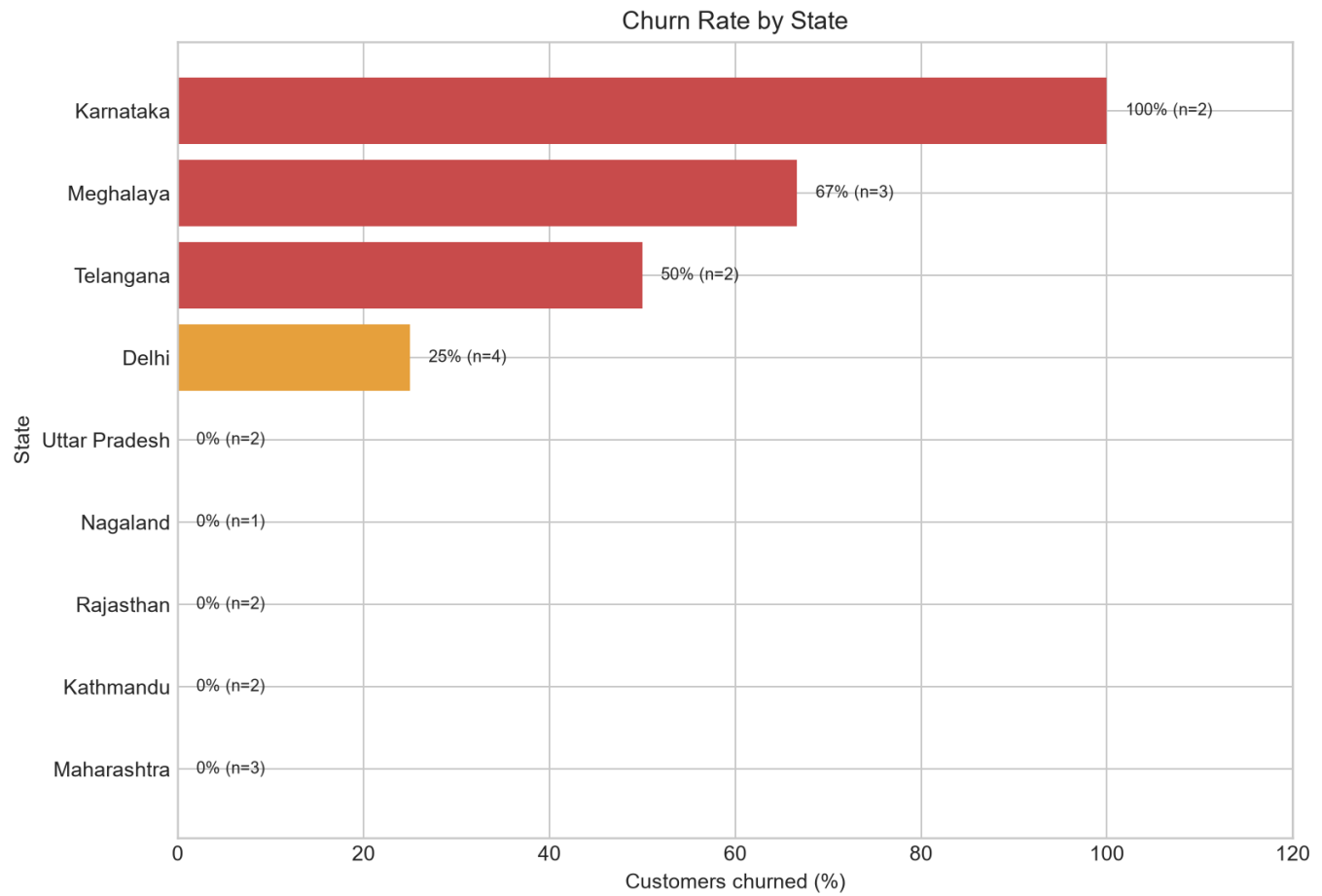
Churn Rate by Plan Type



Plan type	Customers	Churned	Churn rate	Monthly charges	Charges at risk
Basic	5	3	60.00%	52.95	38.97
Standard	9	2	22.22%	123.91	21.98
Premium	7	1	14.29%	218.93	12.99

Basic is the clearest plan-level retention priority, while Premium contributes the largest total monthly charges and should be protected despite its lower churn rate.

Churn by State

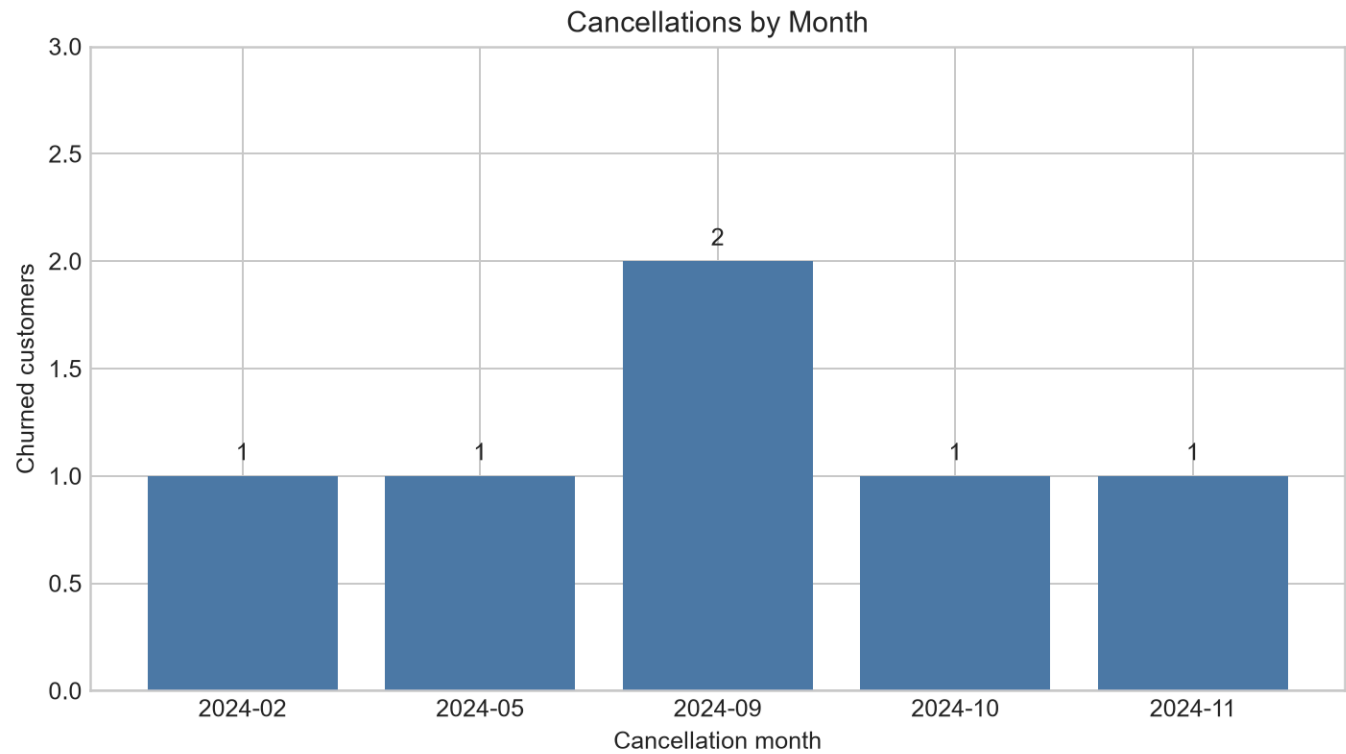


State	Customers	Churned	Churn rate	Monthly charges	Charges at risk
Karnataka	2	2	100.00%	20.98	20.98
Meghalaya	3	2	66.67%	42.97	21.98
Telangana	2	1	50.00%	30.98	16.99
Delhi	4	1	25.00%	52.96	13.99
Kathmandu	2	0	0.00%	20.98	0.00
Maharashtra	3	0	0.00%	50.97	0.00
Nagaland	1	0	0.00%	22.99	0.00
Rajasthan	2	0	0.00%	36.98	0.00
Uttar Pradesh	2	0	0.00%	115.98	0.00

Karnataka's 100% rate is based on only two customers; it should trigger investigation, not a broad regional conclusion. Meghalaya and Telangana also merit a focused review. Uttar Pradesh has the largest total monthly charges (115.98) but no churn in this small sample.

Monthly Churn Trend

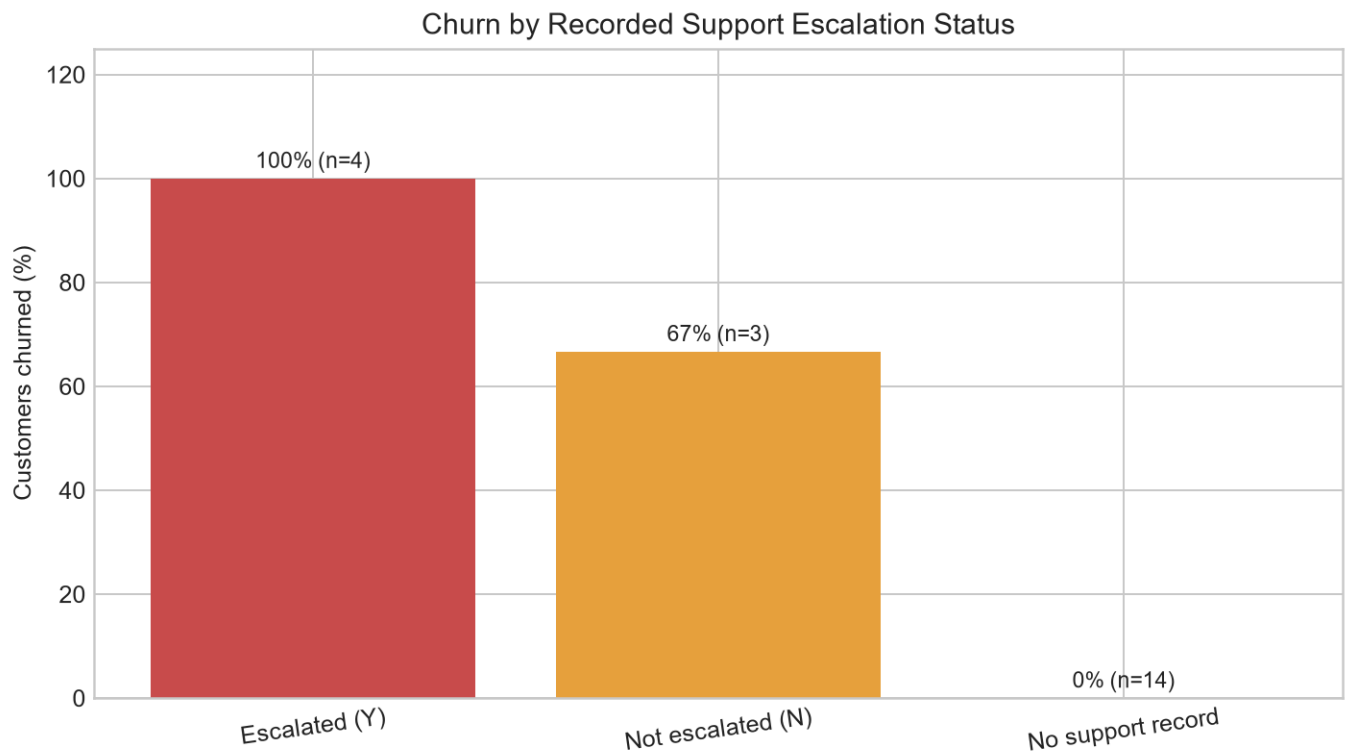
These are **cancellation counts**, not monthly churn rates: customer-base denominators by month are not provided.



Cancellation month (2024)	Churned customers
February	1
May	1
September	2
October	1
November	1

September is the peak cancellation month in the observed period, with two of six cancellations. Missing months in this chart mean no cancellation was recorded in this dataset for that month; they do not establish that the active customer base was stable.

Support Escalations and Churn

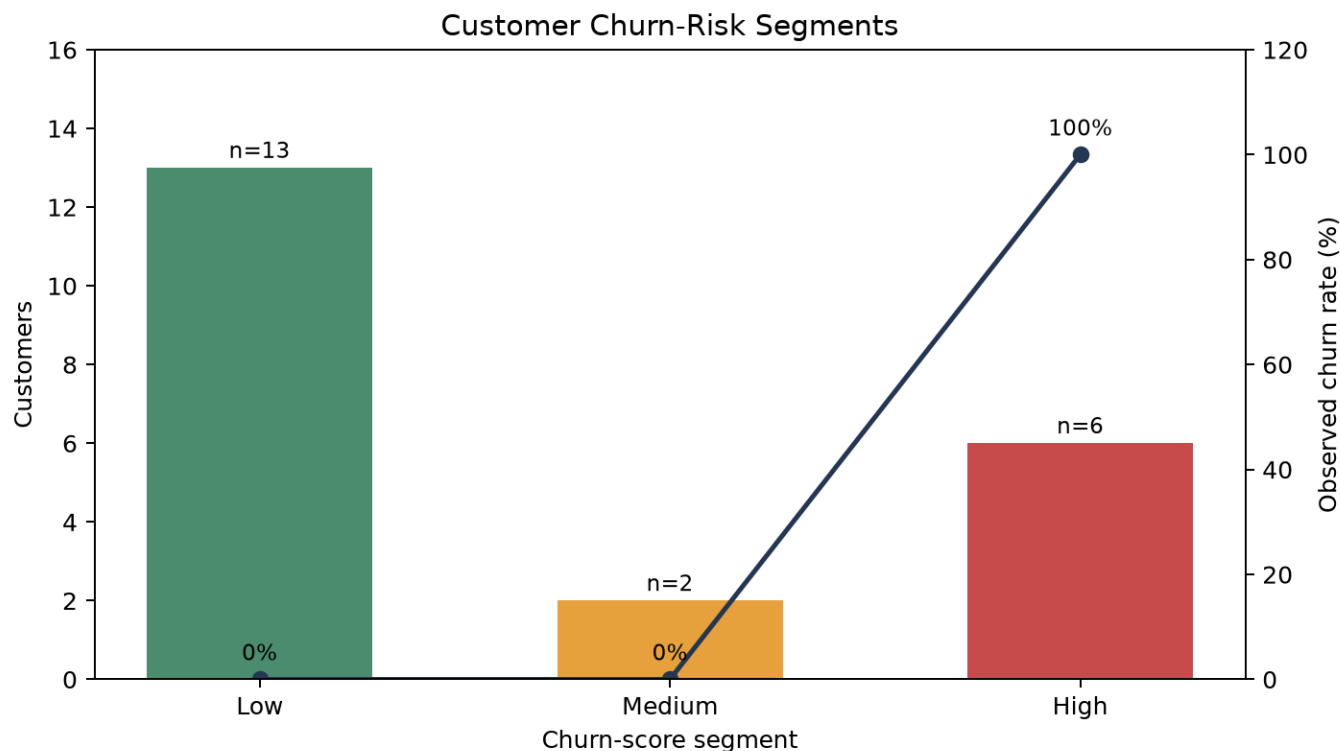


Escalation status	Customers	Churned	Churn rate
Explicit Y	4	4	100.00%
Explicit N	3	2	66.67%
No support record	14	0	0.00%

The four explicitly escalated customers all churned. Among the seven customers with a support record, the Pearson correlation of binary escalation status and churn is **0.47**. The script reports **0.77** because it first converts missing escalation values to 0, then calculates correlation across all 21 customers; that calculation effectively treats absence of a support record as “not escalated.” The 0.77 is therefore not the correlation among customers with observed support data. The chart separates the no-support-record group rather than labeling those customers as not escalated.

This association is descriptive only: there are just seven customers with support records, and the support sample may not be representative. Escalations could be a symptom of an already deteriorating customer experience rather than a cause of churn. The perfect observed churn rate among escalated customers nevertheless supports prompt, human follow-up as a practical retention control.

Churn-Risk Segmentation



Churn-score segment	Score range	Customers	Churned	Churn rate	Monthly charges	Charges at risk
Low	< 50	13	0	0.00%	286.87	0.00
Medium	50–69	2	0	0.00%	34.98	0.00
High	≥ 70	6	6	100.00%	73.94	73.94

All observed churn and all monthly charges at risk fall in the high-score segment. Because the segmentation is evaluated on only 21 records and the score's origin, prediction timing, and validation are unknown, this alignment should not be interpreted as proof of model accuracy or as a prospective guarantee.

Business Insights and Recommendations

- Prioritize Basic-plan retention.** Review cancellation reasons, renewal timing, and service experience for Basic customers. Test a targeted retention offer or plan-fit conversation and compare save rate and margin against a control group.
- Create an escalation recovery workflow.** Route escalated cases to an accountable owner, set a rapid response SLA, and track resolution, post-resolution satisfaction, and subsequent churn. Validate the pattern on a larger period before estimating financial impact.
- Use high churn scores to triage, not to automate adverse decisions.** Contact the six high-score customers proactively, verify the score is available before cancellation, and measure intervention outcomes prospectively.
- Investigate state-level signals with adequate denominators.** Begin with Karnataka, Meghalaya, and Telangana, but collect more customers and examine plan mix, support access, and service quality before allocating region-specific spend.
- Understand competitor and price losses.** Among six cancellation reasons, two customers switched to a competitor and one cited price. Test competitive positioning and affordability options; the remaining reasons were content, streaming quality, and forgotten trial cancellation.

6. **Improve analytics data coverage.** Capture currency/charge units, support-contact coverage, complaint histories, active customer counts by month, and consistent geography. Preserve all complaint events for service analysis rather than relying only on the latest event.
7. **Monitor cohort-based outcomes.** Track monthly churn rate with an eligible-customer denominator, plan/state churn with confidence intervals or minimum sample-size thresholds, recurring revenue lost, and retention campaign lift.

Limitations

- Only 21 customers and six churn events are available; segment and geography percentages are volatile.
- This is not a time-series customer-base panel, so monthly cancellation counts cannot be read as monthly churn rates.
- Currency and scale for `monthly_charges` are undocumented; revenue figures use the raw numeric units.
- Support records exist for only 7 of 21 customers. Missing support is not evidence that a customer had no support interaction.
- The churn score's provenance and point-in-time validity are unknown. Its apparent match with churn requires independent, time-aware validation.
- Correlation and segment summaries do not establish causation.